

NAME:_____

Math 227C Homework 3

Spring 2026

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1. Consider the weather Markov chain we discussed in class, with two states, state 1 aka rainy, and state 2 aka sunny.
 - a) Write down the transition probability matrix P . Given the shape of the graph associated to the chain, can you conclude that this chain has a unique stationary distribution? Explain.
 - b) Show directly and by hand that $\lambda = 1$ is an eigenvalue, and that P has a unique eigenvector π with eigenvalue 1 and such that $\pi_1 + \pi_2 = 1$.
 - c) Calculate numerically the probability that it is rainy at time $t = 3$, assuming that it is sunny at time $t = 0$.
 - d) Choose an arbitrary initial condition π^0 , and iterate the system several times to illustrate the convergence towards the unique stationary distribution π .
2. Consider the casino problem discussed in class, where a gambler successively bets one dollar to either win another dollar or lose one dollar, until the gambler reaches either N dollars or zero dollars.
 - a) Write code that implements simulations of this process by starting with an initial amount, and successively betting until the gambler stops. Illustrate this code by showing graphs of sample runs.
 - b) Estimate numerically the likelihood of winning at this game as a function of the initial dollar amount. Show a graph with this likelihood for $N = 10$, $p = 1/2$ as a function of initial amounts from zero to N .
3. Consider the discrete time, continuous space system

$$Z_t = \alpha Z_{t-1} + \beta + \epsilon_t,$$

where $\epsilon_t \sim N(0, \sigma^2)$ and $\alpha = 1/2$, $\beta = 4$, $\sigma = 2$.

- a) Write down a formula for the density of Z_t given that $Z_{t-1} = z_0$.
- b) Write the equation of the density π^t as a function of π^{t-1} , and what equation the stationary distribution π satisfies.
- c) Using your preferred programming language, run a numerical simulation of this system in order to observe the evolution of the distributions over time. Start with a sample of points that are uniformly distributed from 0 to 1 as your initial distribution. Then use the formula for time evolution in order to calculate multiple independent samples Z_t over a fixed number of iterations. Show a graph of the estimated numerical distribution of Z_t for two different times.

- d) Using the code from c), show whether the distribution π^t converges towards a stationary distribution π , and plot this estimated distribution numerically.
4. Calculate the Fokker-Planck equation for a Brownian motion $dX = \sigma dB$ with constant variance σ . Show that $N(0, \sigma^2 t)$ is a solution of this equation for the density function $p(x, t)$.
5. Implement the basic Brownian motion construction starting with fixed values of $\Delta X > 0$, $\Delta t > 0$.
- a) Starting with $\Delta X = 1/2$, $\Delta t = 1/3$, write code that simulates $X(t)$ for $t = 0, \Delta t, 2\Delta t, \dots, 10$. Illustrate this code with a graph of multiple sample runs of $X(t)$.
- b) Tie the two increments together with the formula $\Delta x = \sigma\sqrt{\Delta t}$, $\sigma = 2$. Draw two more graphs of sample functions $X(t)$ for increasingly small values of Δt .
- c) Show a numerical distribution of $X(10)$, and compare it with the theoretical Gaussian distribution, for a small value of Δt .